Practice

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Examples 1, 3, and 4

Solve each equation. Check your solution.

1. $7c + 12 = -4c + 78$

2. $2m - 13 = -8m + 27$

3. $9x - 4 = 2x + 3$

4. $6 + 3t = 8t - 14$

5. $\frac{b-4}{6} = \frac{b}{2}$

6. $\frac{3v+12}{6} = \frac{4v}{3}$

7. $2(r + 6) = 4(r + 4)$

8. $6(n + 5) = 3(n + 16)$

9. $5(g + 8) - 7 = 117 - g$

10. $12 - \frac{4}{5}(x + 15) = \left(\frac{2}{5}x + 6\right)$

11. $3(3m - 2) = 2(3m + 3)$

12. $6(3a + 1) - 30 = 3(2a - 4)$

13. $7n + 6 = 4n - 9$

14. $-6(2r + 8) = -10(r - 3)$

15. $5 - 3(w + 4) = w - 7$

16. $2x - 5(x - 3) = 2(x - 10)$

Example 2

17. OLYMPICS In the 2010 Winter Olympic Games, the United States won 1 more than 4 times the number of gold medals France won. The United States won 7 more gold metals than France. Write and solve an equation to find the number of gold medals each country won.

18. REASONING Diego's sister is twice his age minus 9 years. She is also as old as half the sum of the ages of Diego and both of his 12-year-old twin brothers. Write and solve an equation to find the ages of Diego and his sister.

19. NATURE The table shows the current heights and average growth rates of two different species of trees. Write and solve an equation to find how long it will take for the two trees to be the same height.

Tree Species	Current Height	Annual growth
A	38 inches	4 inches
B	45.5 inches	2.5 inches

20. WEIGHT A dog weighs two pounds less than three times the weight of a cat. The dog also weighs twenty-two more pounds than the cat. Write and solve an equation to find the weights of the dog and the cat.

Example 4 Use Literal Equations

PARTY **The total amount of money Kishi spends on pizza for a party is $T = 13.49c + 15.49p$. T represents the total amount of money she spends, c represents the number of cheese pizzas she buys, and p represents the number of pepperoni pizzas she buys.**



Part A

If Kishi has $85 to spend on pizza, describe the constraints on $T = 13.49c + 15.49p$.

- The maximum number of cheese pizzas Kishi can buy is 6, because she can buy 6 cheese pizzas and no pepperoni pizzas without exceeding $85.
- The maximum number of pepperoni pizzas Kishi can buy is 5, because she can buy 5 pepperoni and no cheese pizzas without going over her budget.
- The minimum number of each type of pizza she can buy is 0, because you cannot buy a negative number of pizzas.

Part B

Solve $T = 13.49c + 15.49p$ for c.

$T = 13.49c + 15.49p$	Original equation
$T - 15.49p = 13.49c + 15.49p - 15.49p$	Subtract $15.49p$.
$T - 15.49p = 13.49c$	Simplify.
$\frac{T - 15.49p}{13.49} = \frac{13.49c}{13.49}$	Divide by 13.49.
$\frac{T - 15.49p}{13.49} = c$	Simplify.

Think About It!

Why did you round your answer in Part C to 2 instead of 3?

Part C

If Kishi has $85 to spend on pizza and she needs to buy 3 pepperoni pizzas, find the maximum number of cheese pizzas she can buy.

$c = \frac{T - 15.49p}{13.49}$	Original equation solved for c
$c = \frac{85 - 15.49(3)}{13.49}$	$T = 85$, $p = 3$
$c = \frac{38.53}{13.49}$	Simplify.
$c \approx 2.86$	Divide.

Kishi can buy a maximum of 2 cheese pizzas.

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Example 4

23. **COFFEE SHOP** Consuelo is buying flavored coffee and plain coffee. The total amount of money she spends on coffee is $T = 5.50p + 7f$, where p represents the cost of a package of plain coffee and f represents the cost of a package of flavored coffee.

a. If Consuelo has $40 to spend on coffee, describe the constraints on the formula.

b. Solve for f.

c. If Consuelo needs to buy 3 packages of plain coffee, what is the maximum number of packages of flavored coffee she can buy?

24. **SHOPPING** Kimberly is ordering bath towels and washcloths for the inn where she works. The total cost of the order is $T = 6b + 2w$, where T is the total cost, b is the number of bath towels, and w is the number of washcloths.

a. Her budget is $85. Describe the constraints.

b. Solve for b.

c. She needs to order at least 20 washcloths. How many bath towels can she order and stay under budget?

Examples 5–7

25. **ENVIRONMENT** The United States released 5.877 billion metric tons of carbon dioxide into the environment through the burning of fossil fuels in a recent year. If 1 trillion pounds = 0.4536 billion metric tons, how many trillion pounds of carbon dioxide did the United States release in that year?

26. **EUROS** Trent purchases 44 euros worth of souvenirs while on vacation in France. If $1 U.S. = 0.678 euros, find the cost of the souvenirs in United States dollars.

27. **LENGTH** A pencil is 13.5 centimeters long. If 1 centimeter = 0.39 inch, what is the length of the pencil in feet, to the nearest hundredth?

28. **TRACK** If a track is 400 meters around, how many laps around the track would it take to run 3.1 miles? Round to the nearest tenth. (*Hint:* 1 foot = 0.3048 meter)

Example 5

Graph each equation by using the *x*-and *y*-intercepts.

9. $y = 4 + 2x$

10. $5 - y = -3x$

11. $x = 5y + 5$

12. $x + y = 4$

13. $x - y = -3$

14. $y = 8 - 6x$

Example 6

15. SCHOOL LUNCH Amanda has $210 in her school lunch account. She spends $35 each week on school lunches. The equation $y = 210 - 35x$ represents the total amount in Amanda's school lunch account y for x weeks of purchasing lunches.

a. Find the *x*- and *y*-intercepts and interpret their meaning in the context of the situation.

b. Graph the equation by using the intercepts.

16. SHIPPING The *OOCL Shenzhen*, one of the world's largest container ships, carries 8063 TEUs (1280-cubic-feet containers). Workers can unload a ship at a rate of 1 TEU every minute. The equation $y = 8063 - 60x$ represents the number of TEUs on the ship y after x hours of the workers unloading the containers from the *Shenzhen*.

a. Find the *x*- and *y*-intercepts and interpret their meaning in the context of the situation.

b. Graph the equation by using the intercepts.

Example 4

23. **SPORTS** Wanda is the manager for the soccer team. One of her duties is to hand out cups of water at practice. Each cup of water is 4 ounces. She begins practice with a 128-ounce cooler of water.

a. Create a function to represent the arithmetic sequence.

b. Graph the function.

c. How much water is remaining after Wanda hands out the 14th cup?

24. **THEATER** A theater has 20 seats in the first row, 22 in the second row, 24 in the third row, and so on for 25 rows.

a. Create a function to represent the arithmetic sequence.

b. Graph the function.

c. How many seats are in the last row?

25. **POSTAGE** The price to send a large envelope first class mail is 88 cents for the first ounce and 17 cents for each additional ounce. The table shows the cost for weights up to 5 ounces.

Weight (ounces)	1	2	3	4	5
Postage (dollars)	0.88	1.05	1.22	1.39	1.56

Source: United States Postal Service

a. Create a function to represent the arithmetic sequence.

b. Graph the function.

c. How much did a large envelope weigh that cost $2.07 to send?

26. **VIDEO DOWNLOADING** Brian is downloading episodes of his favorite TV show to play on his personal media device. The cost to download 1 episode is $1.99. The cost to download 2 episodes is $3.98. The cost to download 3 episodes is $5.97.

a. Create a function to represent the arithmetic sequence.

b. Graph the function.

c. What is the cost to download 9 episodes?

Steps 2, 3 Write the equation of the parallel line.

Use the point-slope form to rewrite the equation in slope-intercept form.

$y - y_1 = m(x - x_1)$	Point-slope form
$y - 2 = 3[x - (-4)]$	$(x_1, y_1) = (-4, 2)$ and $m = 3$
$y - 2 = 3(x + 4)$	Simplify.
$y - 2 = 3x + 12$	Distributive Property
$y = 3x + 14$	Add 2 to each side.

Check

Write an equation for the line that passes through (8, 2) and is parallel to the graph of $y = \frac{3}{4}x + 2$.

Study Tip

Checking Your Work
To check that your equation represents the correct line, graph both lines. Verify that the lines appear to be parallel and that your line passes through the given point.

Example 8 Perpendicular Line Through a Given Point

Write an equation in slope-intercept form for the line that passes through (1, −2) and is perpendicular to the graph of $3x + 2y = 12$.

Step 1 Identify the slope of the given line.

Write the equation in slope-intercept form.

$3x + 2y = 12$	Original equation
$3x - 3x + 2y = 12 - 3x$	Subtract $3x$ from each side.
$2y = -3x + 12$	Simplify.
$\frac{2y}{2} = \frac{-3x}{2} + \frac{12}{2}$	Divide each side by 2.
$y = -\frac{3}{2}x + 6$	Simplify.

The slope of the line with equation $3x + 2y = 12$ is $-\frac{3}{2}$. The slope of the line perpendicular to that line is the opposite reciprocal, $\frac{2}{3}$.

Steps 2, 3 Write the equation of the perpendicular line.

Use the point-slope form to rewrite the equation in slope-intercept form.

$y - y_1 = m(x - x_1)$	Point-slope form
$y - (-2) = \frac{2}{3}(x - 1)$	$(x_1, y_1) = (1, -2)$ and $m = \frac{2}{3}$
$y + 2 = \frac{2}{3}(x - 1)$	Simplify.
$y + 2 = \frac{2}{3}x - \frac{2}{3}$	Distributive Property
$y = \frac{2}{3}x - \frac{8}{3}$	Subtract 2 from each side.

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Check

Select the equation in slope-intercept form for the line that passes through (5, 0) and is perpendicular to the graph of $x - 6y = 1$.

A. $y = -6x + 30$

B. $6x + y = 30$

C. $y = -\frac{1}{6}x + 2$

D. $x - 6y = 5$

Example 9 Determine Line Relationships

Determine whether $\overleftrightarrow{AB}$ and $\overleftrightarrow{EF}$ are *parallel*, *perpendicular*, or *neither* for *A*(6, 8), *B*(2, 5), *E*(−6, −3), and *F*(0, 5).

Step 1 Find the slope of each line.

slope of $\overleftrightarrow{AB} = \frac{8-5}{6-2} = \frac{3}{4}$ slope of $\overleftrightarrow{EF} = \frac{-3-5}{-6-0} = \frac{-8}{-6}$ or $\frac{4}{3}$

Step 2 Determine the relationship.

parallel To determine whether the lines are parallel, compare their slopes. The two lines do not have the same slope, so they are not parallel.

perpendicular To determine whether the lines are perpendicular, find the product of their slopes. $\frac{3}{4} \cdot \frac{4}{3} = 1$

Since the product of the slopes is not −1, $\overleftrightarrow{AB}$ and $\overleftrightarrow{EF}$ are not perpendicular.

Check

Determine whether $\overleftrightarrow{CD}$ and $\overleftrightarrow{KL}$ are *parallel*, *perpendicular*, or *neither* for *C*(4, 10), *D*(−1, 12), *K*(6, −5), and *L*(1, −3).

$\overleftrightarrow{CD}$ and $\overleftrightarrow{KL}$ are ? .

Complete each sentence given $y = ax - 5$ and $y = bx + 3$.

When $a = 4$ and $b = 4$, the graphs are ? .

When $a = -3$ and $b = 5$, the graphs are ? .

When $a = -2$ and $b = \frac{1}{2}$, the graphs are ? .

Go Online to practice what you've learned about writing linear equations in the Put It All Together over Lessons 5-1 and 5-2.

Go Online You can complete an Extra Example online.